

UNIVERSITY OF ALBERTA

PSYCH 213 - Spring 2026

Course Project II:

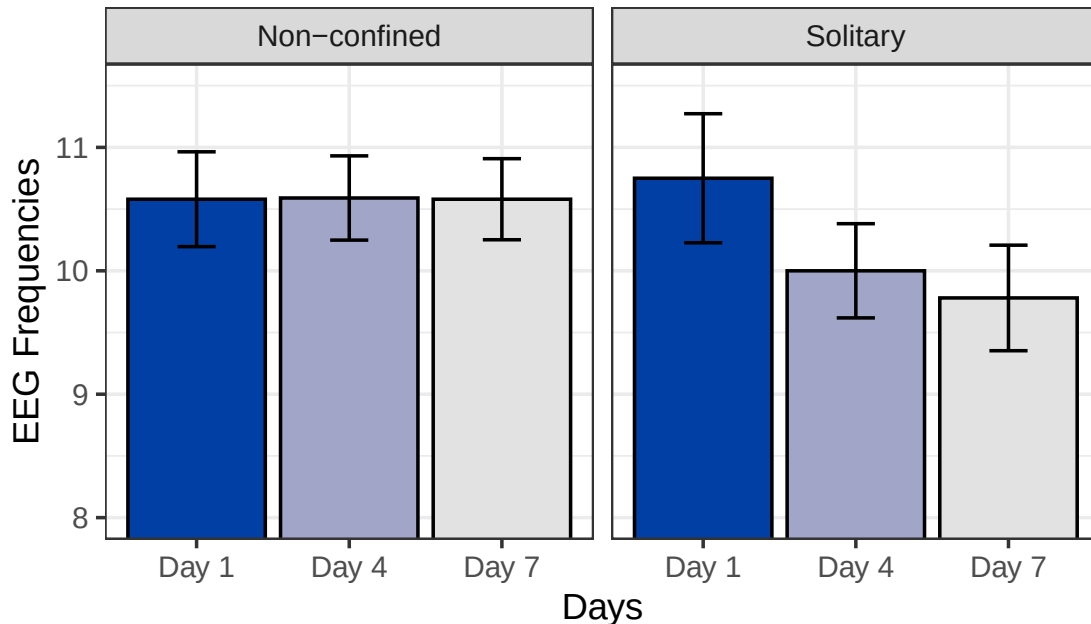
Visualizing Reported Data from Gendreau et al. (1972) Using R

Instructor: **Dr. Jeffrey M. Pisklak**

Objective

Create a Google Colab notebook that performs the following two tasks:

1. **Reproduce the bar graph shown below exactly.**



Here are some points to note about the graph to help you:

- The graph displays means and 95% confidence intervals from the data file `gendreau_et_al_1972.csv`.
 - These are the same type of confidence intervals you used when you learned about one-sample *t*-tests.
 - *Tip*: The data file is in a wide format. Your work will be much easier if you first convert it to a “tidy” (also known as the “long”) format.
- R has built-in colour palettes. This graph uses an HCL-based palette named “Blues 2”. To access HCL palettes, use the `hcl.colors()` function.
 - You will need to determine how to apply this specific colour palette to your graph (we will not cover this in class).
 - *Helpful resource*: <https://journal.r-project.org/articles/RJ-2023-071/>
- ggplot2 default theme was changed, but everything else (e.g., font sizes, font colours, etc.) remain at their default values.
- The graph’s dimensions are 12 cm × 7 cm.

2. Display a data frame of the summary statistics used in the graph.

Your notebook must generate and display a data frame containing the means and 95% confidence intervals shown in the bar graph. The structure of your data frame should resemble the example table below and the numeric values should match (you do not need to round to 4 decimal places).

Table 1: An example of the data frame your code should display

group	days	m	ci_low	ci_top
Non-confined	day_1	10.58	10.1958	10.9642
Non-confined	day_4	10.59	10.2487	10.9313
Non-confined	day_7	10.58	10.2517	10.9083
Solitary	day_1	10.75	10.2273	11.2727
Solitary	day_4	10.00	9.6185	10.3815
Solitary	day_7	9.78	9.3524	10.2076

Rules and Requirements

- Do **not** edit the data file `gendreau_et_al_1972.csv`. Any modifications must be performed in your R code *after* loading the file.
- The data frame must be calculated directly from the data file. Manually entering numbers into your code is not allowed.
- When your notebook is run, the plot should save as a 12 cm × 7 cm `.PDF` file.
- When your notebook is run, it must execute **without any errors**. If *any* errors occur, the entire assignment receives a 30% reduction on its overall score.
 - Test your notebook by selecting *Runtime > Restart session and run all*.
- The only R package you may use is the `tidyverse`.

Submission Instructions

Submit the following file:

- A copy of your notebook (`.IPYNB`) file.

Due Date

This is due Friday June 5th by 23:59.

About the Data

The data set `gendreau_et_al_1972.csv` contains the results of a study that examined the EEG frequencies of 20 prison inmates participating in a study at the notorious Kingston Penitentiary in Ontario. Half were randomly assigned to a “Solitary” group that placed them in solitary confinement, and the other half were assigned to a “Non-confined” control group. Measurements were taken at 1, 4, and 7 days.

Gendreau, P., Freedman, N. L., Wilde, G. J., & Scott, G. D. (1972). Changes in EEG alpha frequency and evoked response latency during solitary confinement. *Journal of abnormal psychology*, 79(1), 54–59.
<https://doi.org/10.1037/h0032339>